



Pet Adoption and Rescue Management Portal

A centralized web-based ecosystem for reuniting lost pets with their owners.



THE HUMAN CONTEXT

The Reality of Loss in Modern Cities



Global Scale

Pets go missing across cities worldwide every day.



The Struggle

Owners face immense difficulty locating pets quickly due to a lack of coordination.



Emotional Toll

The current chaotic landscape causes deep emotional stress for pet owners.

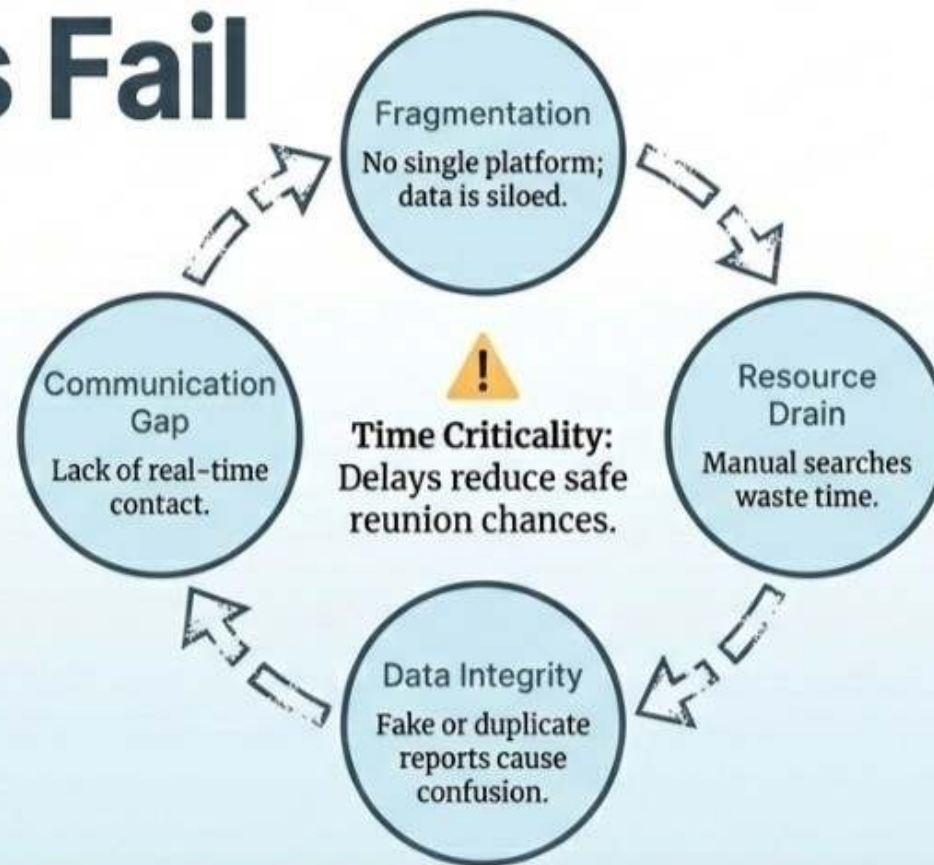


Systemic Failure

Information remains scattered across disparate platforms; no centralized rescue coordination exists.

THE PROBLEM STATEMENT

Why Traditional Recovery Methods Fail



THE SOLUTION

A Unified Rescue Ecosystem

A centralized web-based rescue platform designed for community participation. The system acts as a bridge, connecting owners with pet finders through a user-friendly interface.

- ✓ Supports 'Lost' and 'Found' cases simultaneously
- ✓ Admin-managed verification ensures data quality
- ✓ Real-time connection between parties



Core Functionality & Features

Onboarding



Secure User
Registration and
Login

Reporting



Lost/Found flows
with photo upload

Discovery



Location-based
search & filtering

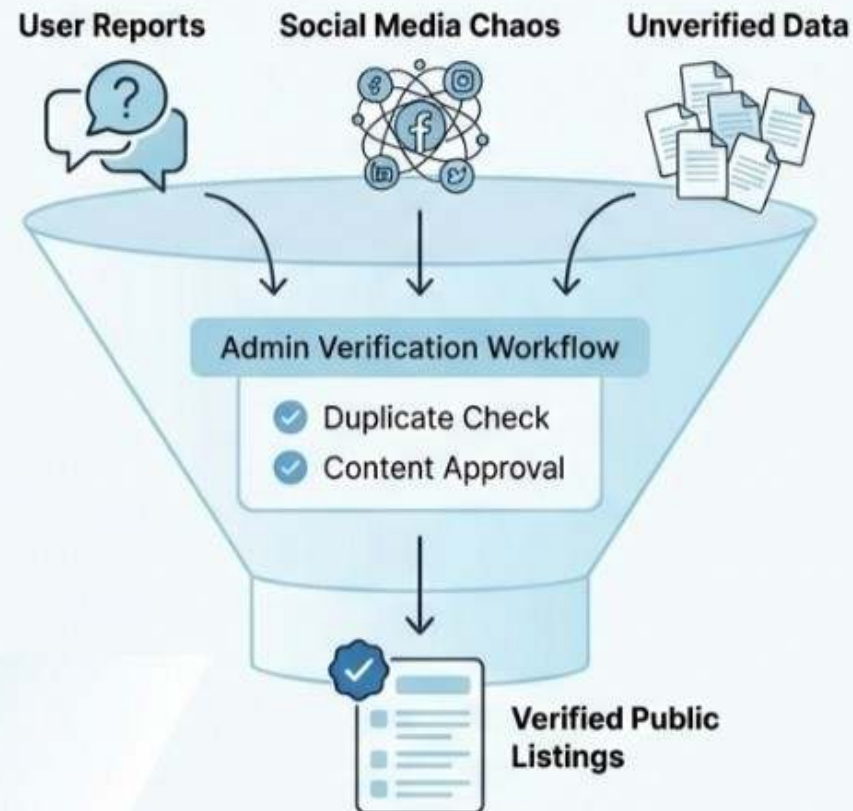
Reunification



Automated email
notifications for
matches

Engineering Trust and Verification

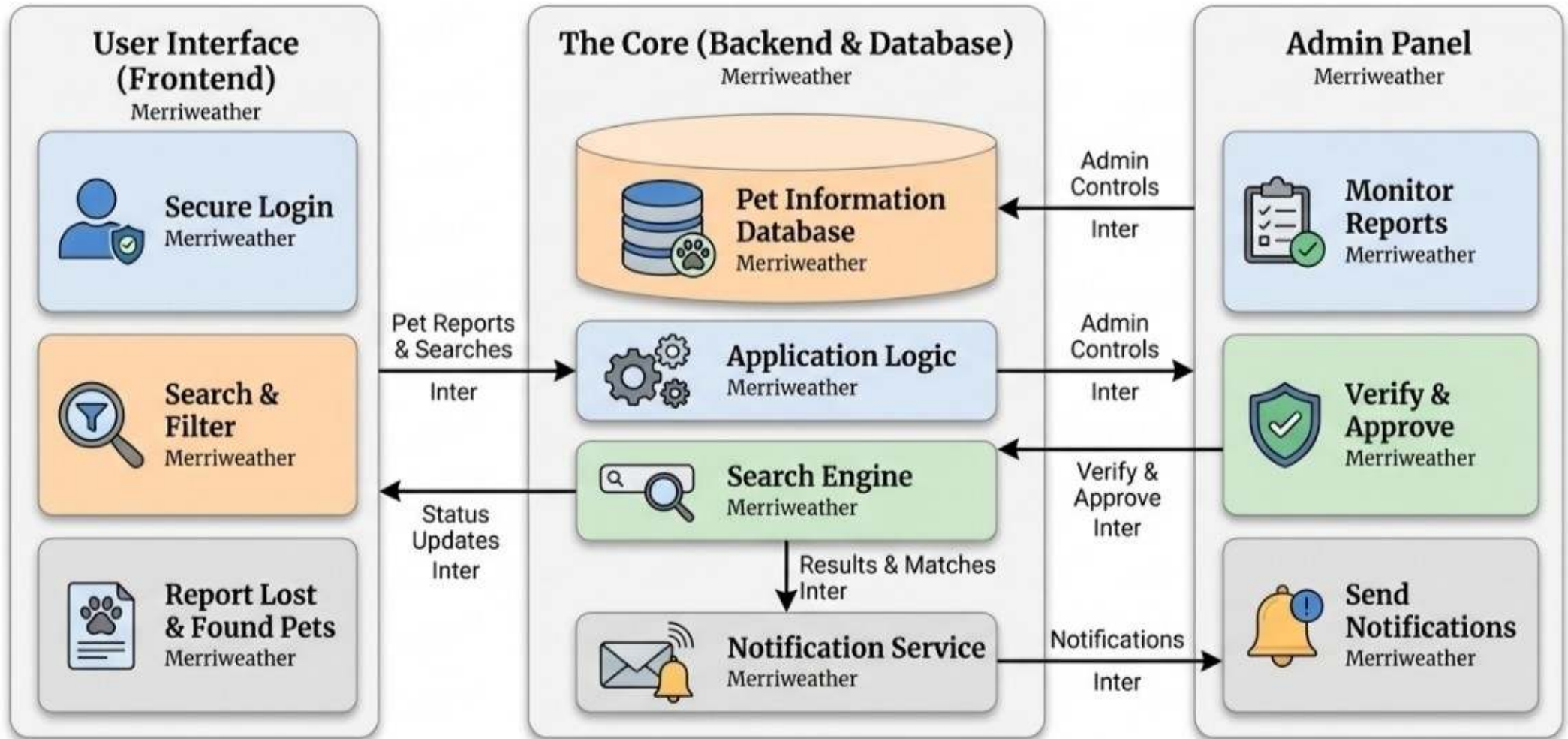
In a landscape of misinformation, we enforce strict data integrity



Safety Features

- ✓ **Secure Communication:** Private channels protecting user data.
- ✓ **Alert System:** Genuine reports trigger instant notifications.

System Architecture



Technology Stack

Frontend

Inter Bootstrap 5

Merriweather
HTML5, CSS3, JavaScript

Responsive Design & AJAX for
async operations.

Backend

Django 4.x

Python 3.9+

Django REST Framework for modular API.

Database

SQL Database Integration

Merriweather

Structured data storage.

Security Implementation



Data Protection

- **Encryption:** Passwords protected via hashing algorithms.
- **Transmission:** HTTPS encryption for all data transfer.

Access Control

- **RBAC:** Role-Based Access Control separating User/Admin privileges.
- **Sessions:** Secure session management with strict timeouts.

Threat Mitigation

- **Validation:** Input sanitization to prevent SQL injection.
- **Defense:** Active protection against common web attacks.

Impact Summary

The Old Way

- ☼ Scattered Information
- Manual Searches
- 🔍 Fake Reports
- High Stress

The New Portal

- ✓ Centralized Ecosystem
- ✓ Location-Based Filtering
- ✓ Admin Verified Data
- ✓ Real-Time Alerts

Transforming chaos into coordination. By leveraging modern web frameworks, we restore not just pets, but peace of mind.

